ShopEase Product Service

FastAPI Product Management Microservice with Docker

Submitted By: Dineshkumar S

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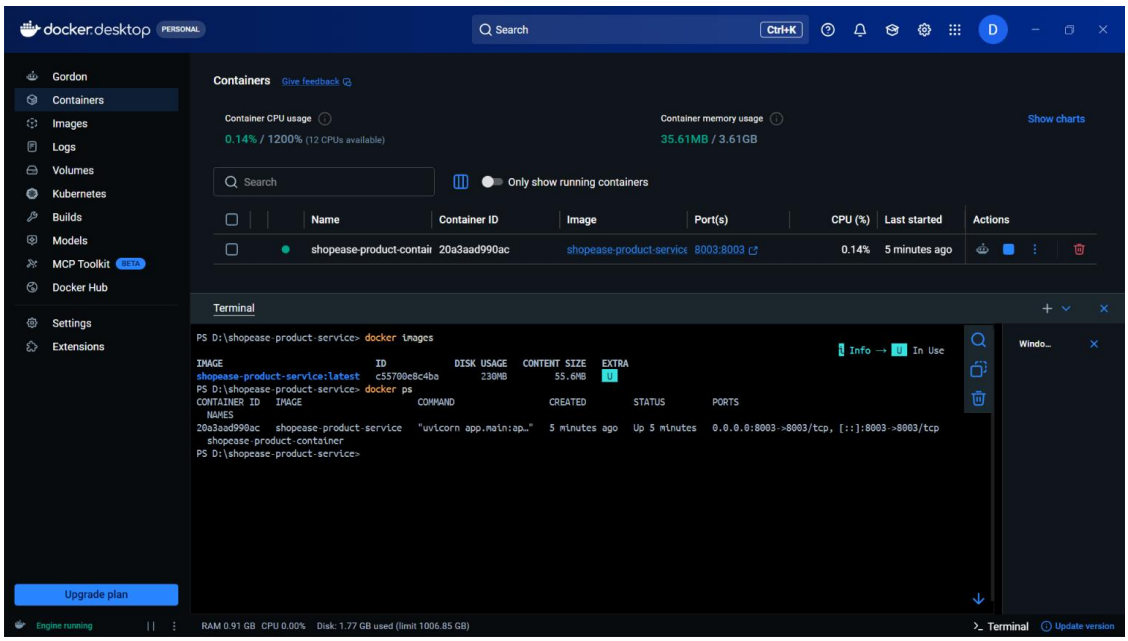
GitHub Repository: https://github.com/Dineshkumarsenthil/shopease-product-service

Requirement Completion

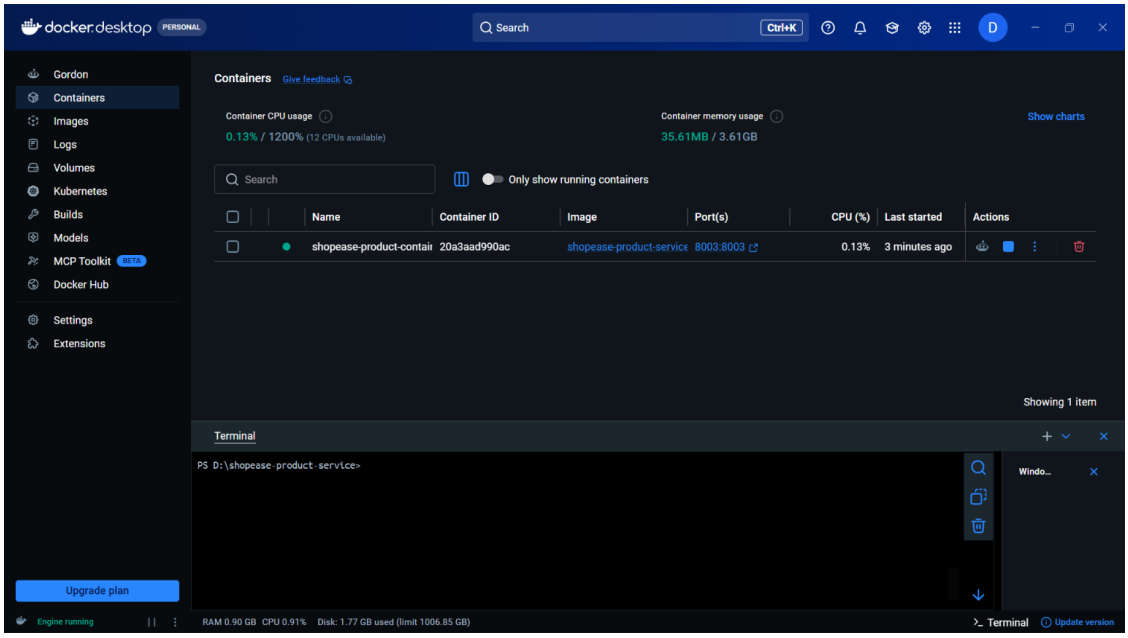
Requirement	Status	Evidence
Product Service Source Code	Done	FastAPI project with separate folders for routers, schemas, models, services and config. main.py only starts the app and registers the router.
Product CRUD APIs	Done	Five endpoints under /api/products: POST, GET (list), GET (by ID), PUT and DELETE. List, category filter, pagination, update and delete were tested in Swagger UI.
In-Memory Storage	Done	Products are stored in a Python dictionary. IDs are generated automatically and each product has a created date.
Pydantic Validation	Done	Schemas check name (required, contains letters), price (greater than 0), stock (0 or more) and status (Active or Inactive). Invalid input returns 422.
Exception Handling	Done	Custom handlers return 404 (not found), 400 (invalid ID), 422 (validation) and 500 (unexpected error) in one JSON format.
Dockerfile	Done	Based on python:3.11-slim, installs requirements.txt, copies the code, exposes port 8003 and starts Uvicorn.
Docker Ignore File	Done	.dockerignore excludes venv, cache files, .git and .env from the image.
Docker Image Build	Done	Image shopease-product-service:latest built successfully (docker images).
Docker Container Running	Done	Container shopease-product-container is Up, port mapping 0.0.0.0:8003->8003 (docker ps and Docker Desktop).
Swagger UI in Docker	Done	Swagger UI opens at http://localhost:8003/docs and lists all endpoints.
README Documentation	Done	Overview, technology stack, folder structure, installation, local and Docker commands, endpoints and testing steps.
GitHub Repository	Done	Source code pushed to https://github.com/Dineshkumarsenthil/shopease-product-service

Evidence

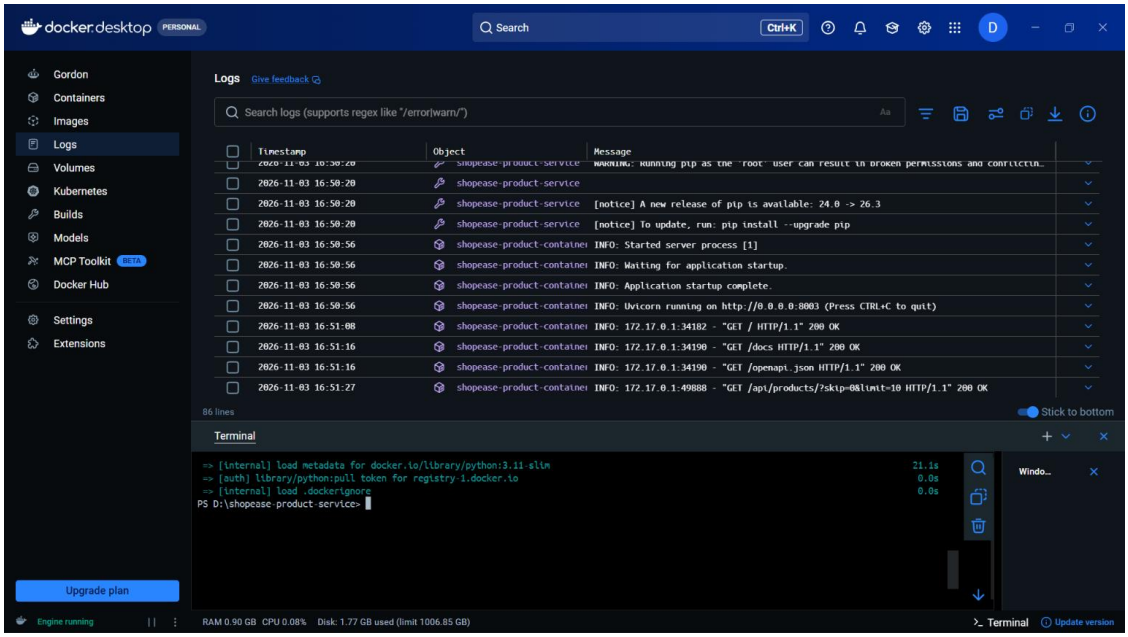
Docker — Image Built and Container Running (docker images, docker ps)



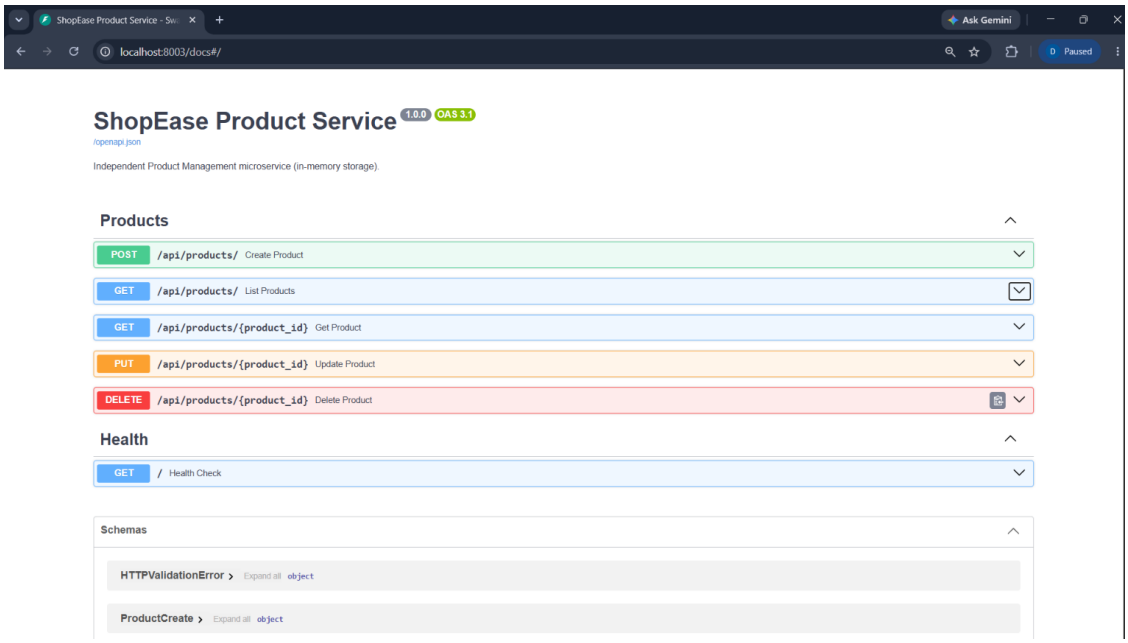
Docker — Container Running in Docker Desktop on Port 8003:8003



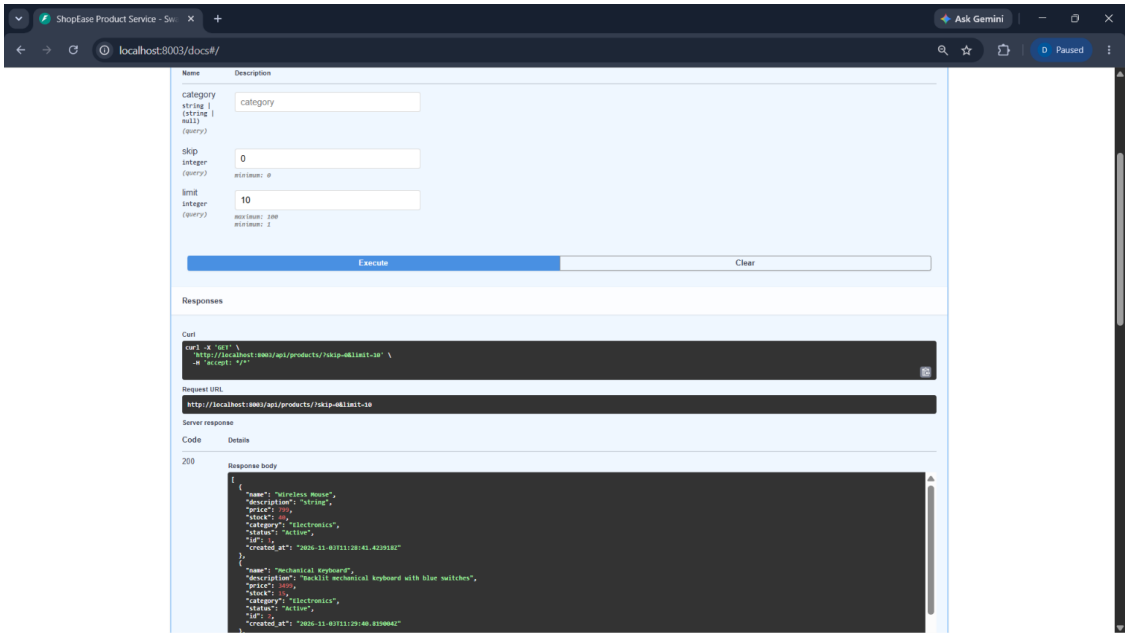
Docker — Container Logs, Uvicorn Running on Port 8003



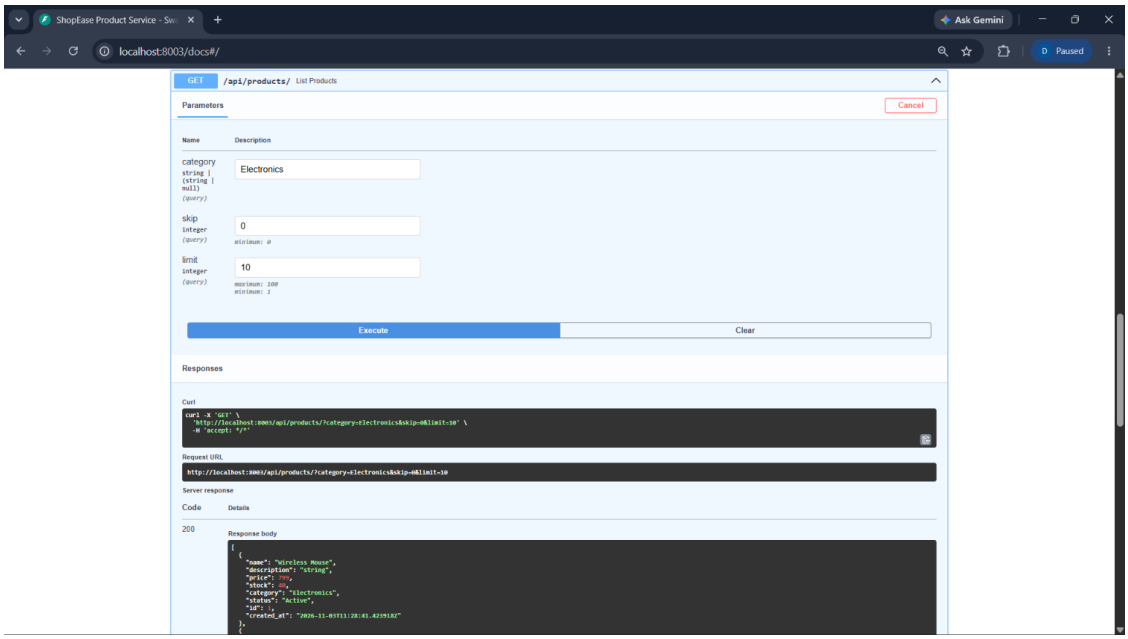
API Documentation — Swagger UI Endpoint List (Served from Docker)



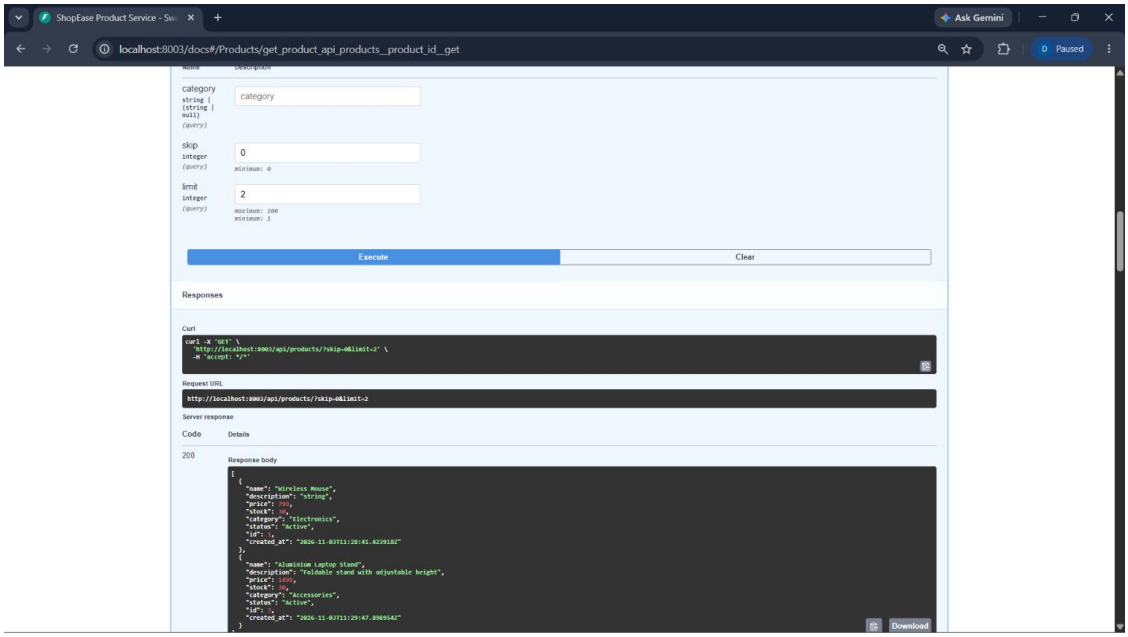
Retrieve All Products — GET /api/products/, 200 OK



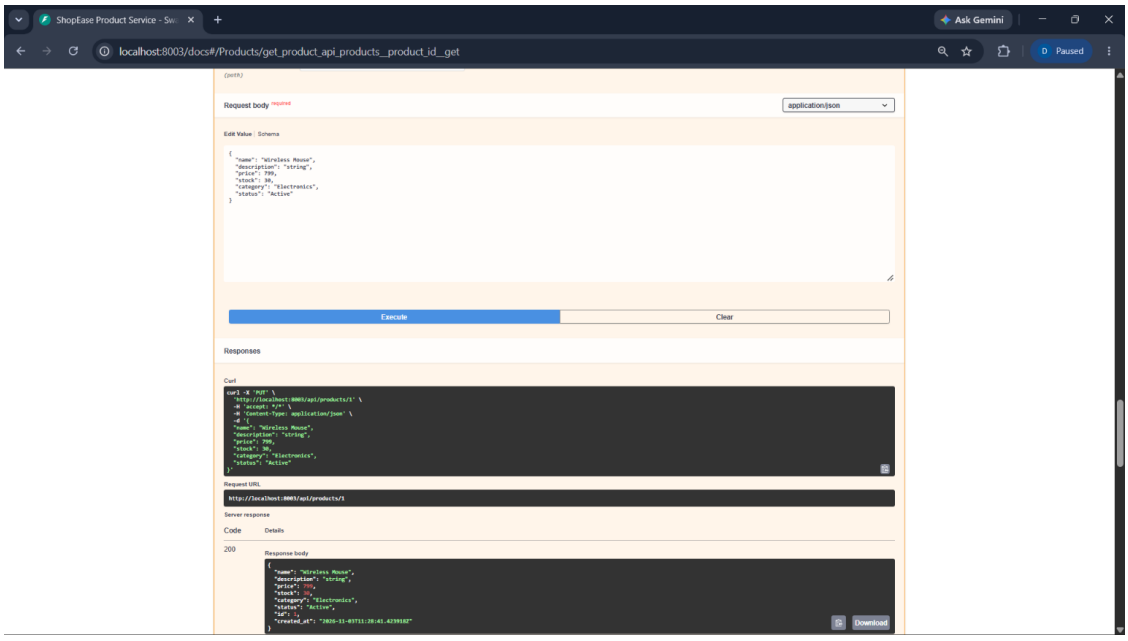
Category Filter — GET /api/products/?category=Electronics, 200 OK



Pagination — GET /api/products/?skip=0&limit=2, 200 OK



Update Product — PUT /api/products/1, 200 OK



Delete Product — DELETE /api/products/3, 204 No Content

